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Paleomagnetism of selected Quaternary sediments on Mt Kenya, East Africa: a reconnaissance study

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Abstract—Samples of Quaternary sediments were collected from the Teleki and Hobley Valley sites on Mt Kenya to determine whether paleomagnetism might assist in placing them in the local time–stratigraphic record. In addition, we sought to determine if paleomagnetic characterization might allow detection of possible errors in radiocarbon dated buried A horizons. Results show that at least some glacial sediments of Mt Kenya were deposited during the Matuyama Polarity Epoch. Late glacial sediments contain a possible paleomagnetic excursion and yield radiocarbon dates ranging from approximately 21,000 to 25,000 yr B.P.

INTRODUCTION

SEDIMENTS IN Hobley (Site HOB19) and Teleki (Site TV61) valleys were described and dated by use of relative (RD) and absolute (AD) dating methods (Mahaney, Piegat, Halvorson and Sanmugadas, 1984), including superposition, relative weathering states for surface boulders, paleosol characteristics which shed light on degree of weathering, scanning electron microscopic study of fine and very fine sand separates (Vortisch and Mahaney, in press) radiocarbon, and amino acid ratios (especially ratios of the D to L-forms of aspartic acid (Mahaney, Boyer and Rutter, 1985). Since various AD and RD parameters suggested ages in the $>2.0 \times 10^4$ to $\sim 1.0 \times 10^5$ and $\sim 5.0 \times 10^5$ to $\sim 1.0 \times 10^6$ yr B.P. range, we sampled two sections for paleomagnetic characterization to determine if paleomagnetic signals would assist in placing each of these sedimentary units more firmly in the time stratigraphic record.

FIELD AREA

Mount Kenya (Fig. 1) is one of the principal high mountains of East Africa. Following each episode of glaciation geologic deposits consisting of tills, ashes, loesses and glaciofluvial deposits were emplaced either in superposition or as a surface sequence overlying bedrock of volcanic or sedimentary origin. As deposition ceased weathering occurred in exposed surficial materials and paleosols formed in a variety of different materials. Many of these paleosols and deposits have

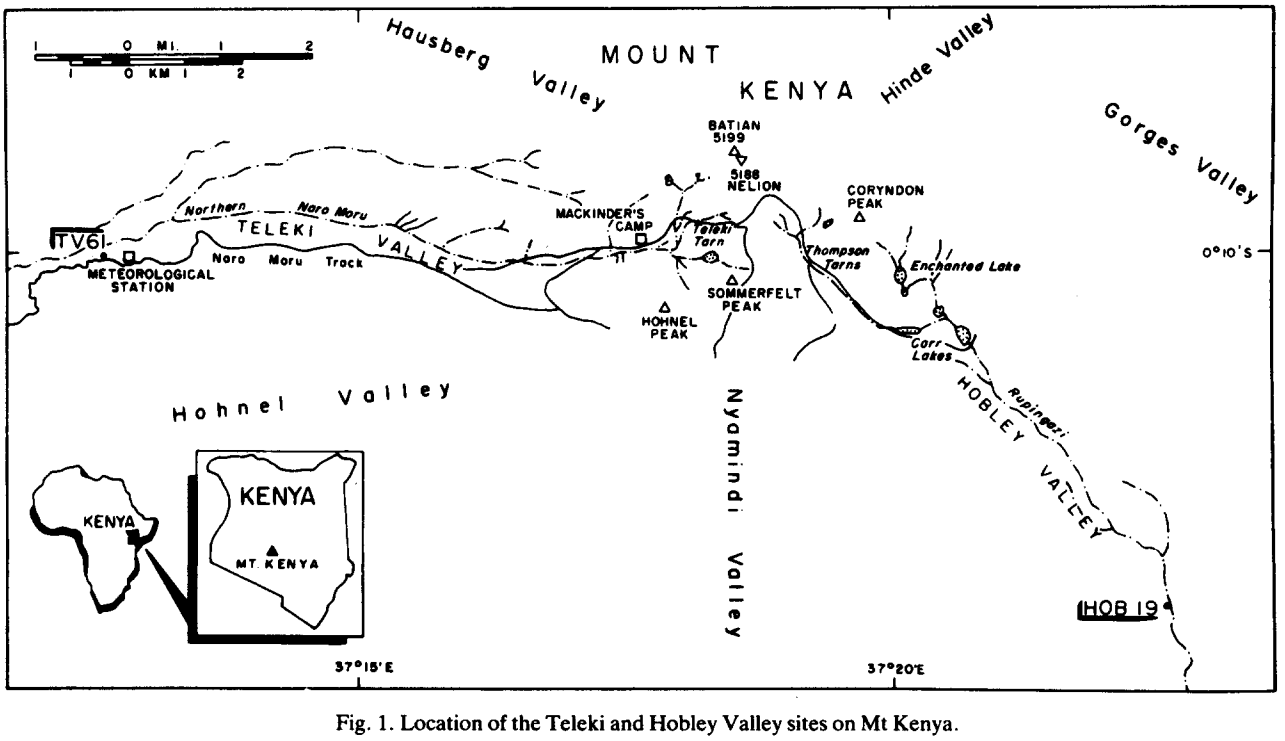


Fig. 1. Location of the Teleki and Hobley Valley sites on Mt Kenya.

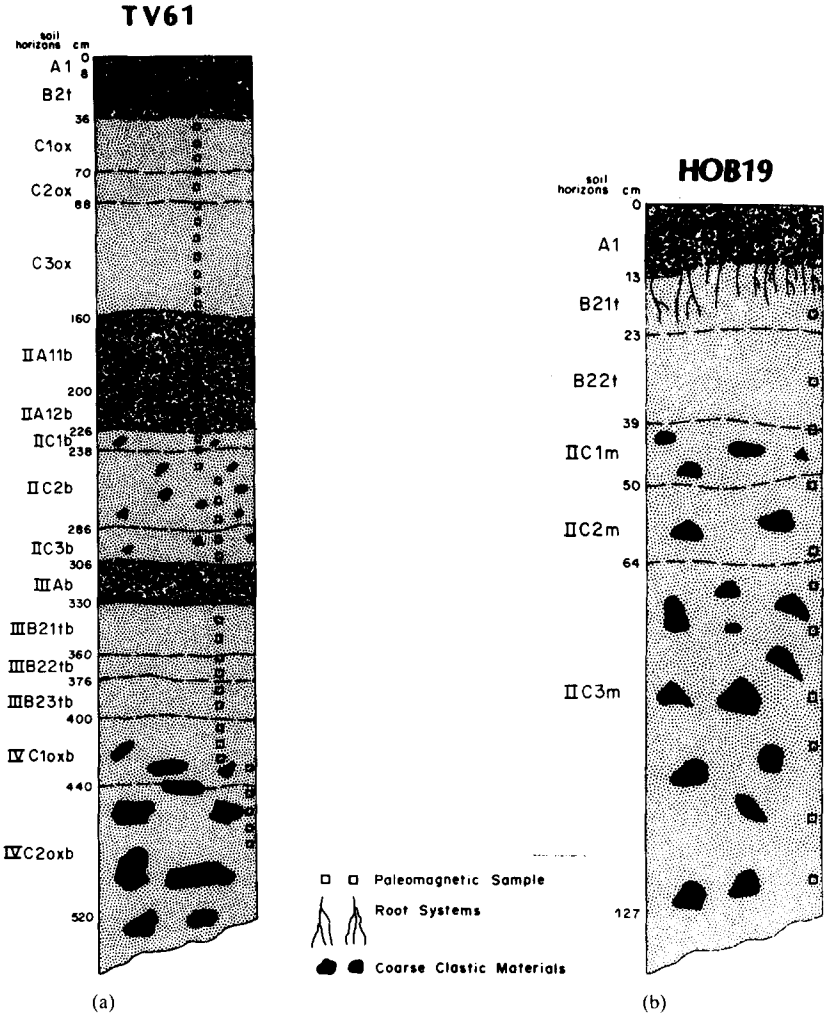


Fig. 2a. Geologic cross-section of site TV61 indicating paleomagnetic sampling locations (Sample 1 at top, 44 at bottom).

Fig. 2b. Geologic cross-section of site HOB19 indicating paleomagnetic sampling locations (Sample 1 at top, 12 at bottom).

been dated (Mahaney, 1979, 1980, 1981, 1982a, 1982b, 1982c, 1983a, 1983b, 1984a, 1984b, 1984c, 1985); however, much additional work needs to be completed in order to firmly place all of them in a time-stratigraphic framework for Mount Kenya.

Climate

The Afroalpine area rises from ~3200 m to the summit of Mount Kenya at 5199 m. Precipitation derived from the Indian Ocean is delivered by the tropical easterly wind belt and the classical monsoon system. The southeast slopes are wettest with rainfall reaching 2500 mm (at 2250 m); the northern slopes are driest with ~1000 mm annually. On the west flank of the mountain, precipitation rises from ~500 mm at Naro Moru (2000 m) to ~1000 mm on the high moorland (~3500 m) and then drops in the central peaks areas (~4300 m) to ~750–800 mm (Coetzee, 1967). Evapotranspiration data are not available for Mount Kenya. Temperature extrapolations (Coe, 1967; Thompson, 1966) suggest that temperatures at 4200 m average 3.1 to 3.6°C (from observations taken during two periods in 1948; Coetzee, 1967; Hedberg, 1964). The drop in precipitation from the western to the northern flanks (Pedgley, 1966) affects leaching effectiveness in the soil systems. This is reflected by differences in soil profile expression along the western, northern and southeastern flanks of the mountain (Mahaney, 1979, 1985).

Vegetation

Above the heather zone at timberline (~3200–3300 m) the lower Afroalpine zone dominated by *Senecio brassica* extends upwards to ~3700 m where it merges with the upper Afroalpine zone characterized by *Senecio keniodendron*. The upper Afroalpine zone merges with vegetation of the nival zone at ~4500 m (Coe, 1967; Hedberg, 1964).

METHODS

Study sites were selected on moraines and slope deposits near the margin of the glaciated area on Mount Kenya. Soil samples were collected from each horizon described in the field (following the nomenclature of the

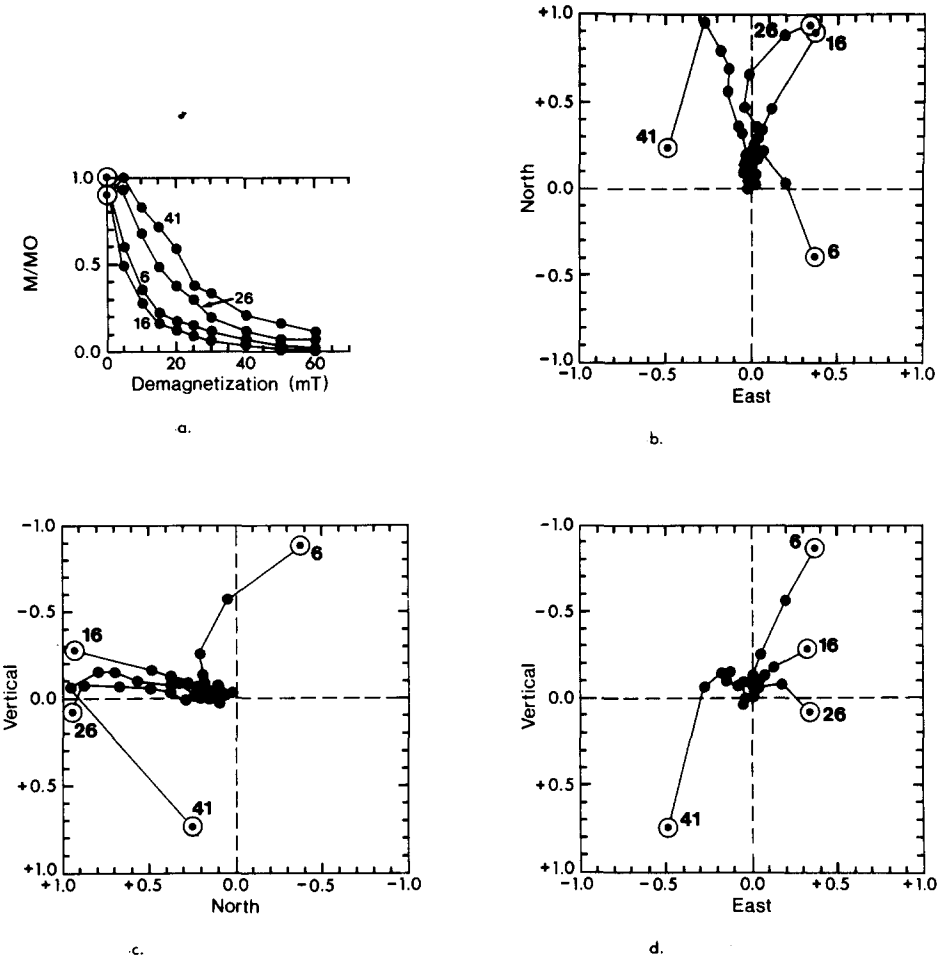


Fig. 3. Typical demagnetization curves of samples 6, 16, 26 and 41 from Teleki Valley. [a] Ratio (M/MO) of natural remanent magnetization (N.R.M.) to residual moment after progressive alternating field demagnetization (a.f.d.) at levels of 5, 10, 15, 20, 25, 30, 40, 50 and 60 millitesla (mT). Magnetization values have been normalized to the highest intensity and are thus unitless. Zijderveld plots showing orthogonal projections of the changing resultant magnetization vector on the horizontal plane [b], on the vertical N–S plane [c], and on the vertical E–W plane [d]. The samples are normally magnetized.

Soil Survey Staff, 1951, 1975; and Birkeland, 1984). Particle size analyses (Folk, 1968) were performed on the <2 mm fractions; coarse grade sizes (>63 μm) were determined by sieving, whereas fine grained sizes (<63 μm) were calculated by sedimentation following procedures established by Bouyoucos (1962) and Day (1965).

Paleomagnetic samples were collected from two sites: the Teleki Valley (TV61) site composed of a complex of three buried soils and a modern surface soil, and the Hobley Valley (HOB19) site composed of till overlain by a thin loess (Fig. 1 for site locations and Figs. 2a and 2b for sampling horizons). After extensive excavation and removal of overburden, undeformed sediments were found which proved suitable for paleomagnetic sampling in 2 × 2 × 2 cm plastic cubes. Azimuthal orientation was made in the field by magnetic compass. The open end of the plastic cube was aligned parallel to the vertical face of the outcrop and samples were taken at 10 cm intervals. Five pilot specimens from the Teleki Valley site and 12 from the Hobley Valley site were selected and these were subjected to stepwise alternating field (a.f.) demagnetization at levels of 5, 10, 15, 20, 25, 30, 40, 50 and 60 millitesla (mT) in a conventional three-axis tumbler. After each demagnetization step, the remanence was measured by means of a Schonstedt DSM-1 magnetometer. Qualitative assessment of changes in magnetic vector directions and magnetization values following a.f. demagnetization led to a choice of 15, 20 and 25 mT as the standard demagnetizing fields for the remaining samples. Typical demagnetization plots are shown in Figs. 3 and 4, and most specimens are characterized by a single component remanent magnetization. On the basis of the low median destructive field of these specimens, it would appear that detrital magnetite is the predominant remanence carrier [magnetite is one of the principal components of most tills on Mount Kenya (Vortisch and Mahaney, 1986)].

DATA AND DISCUSSION

Profile TV61 (Fig. 1 for location; Fig. 2a for geologic cross-section) consists of a multistorey sequence of paleosols in till (IV), in outwash (III) (Paleudalf), in colluvium (II) (Entisol), and in slope wash (ground soil;

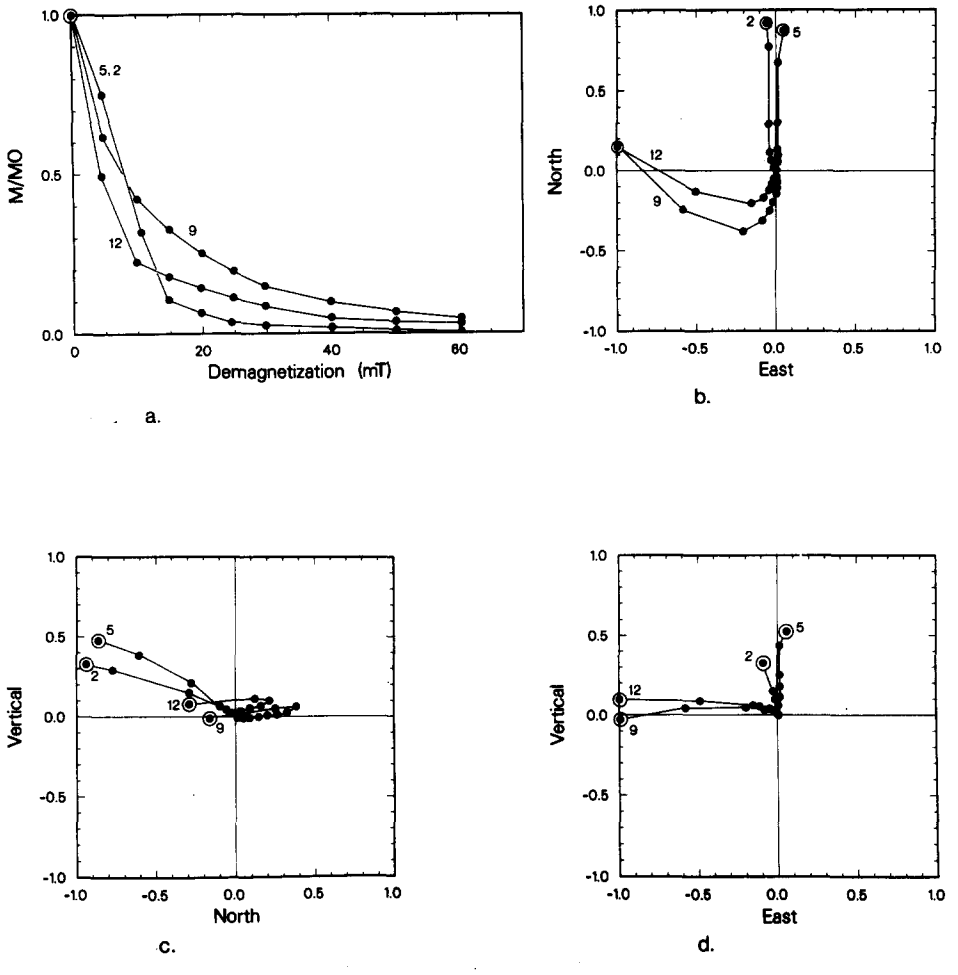


Fig. 4. Typical demagnetization curves of samples 2 and 5 (normal) and samples 9 and 12 (reversed) from Hobley Valley. Description of [a] to [d] as in Fig. 3.

Inceptisol). Horizon differentiation was made primarily on the basis of soil colour and particle size, there being little variation in soil structure which is typical of soils in the Afroalpine and tropical subalpine environments. This paleosol sequence is important because it contains several disconformities, the lowest (III and IV) of these may delineate the upper boundary of the oldest pre-Teleki till (>0.7 × 10^6 yr B.P.) on Mount Kenya.

Profile HOB19 (Fig. 1 for location; Fig. 2b for geologic cross-section) reveals a Paleudalf (Soil Survey Staff, 1975) formed in pre-Teleki till and loess, at 3000 m, in Hobley Valley. The paleosol formed in this sequence is probably several hundred thousand years old (Mahaney, 1983). This profile was chosen for paleomagnetic study because it correlates with, and is representative of, older pre-Telkeki drifts on the eastern side of Mount Kenya (Baker, 1967; Mahaney, 1982a, 1983a). Although these drift surfaces lack distinctive morainal forms, the microfabric (as examined by scanning electron microscopy) shows alignment of very large and large (2000–500 μm) sand grains in the downvalley direction (SE–NW) (personal communication, E. Derbyshire, July, 1985). Such strong orientations suggest emplacement by ice. The paleosol formed in this till is capped with loess that presumably is somewhat younger than the C horizon complex. While not unusually thick (1.1 m for the total pedon), the C horizons have an indurated matrix cemented with Fe_2O_3 and MnO_2 (personal communication, E. Derbyshire, July, 1982).

PALEOMAGNETIC SIGNATURES

Teleki Valley site

The declination, inclination and intensity profiles for all samples are shown in Fig. 5a and the summary of remanent magnetization directions is shown in Table 1. The intensities of the demagnetized samples range from 0.003 to 0.9 AM^{-1} although most samples range from 0.02 to 0.08 AM^{-1}. Undemagnetized intensities are from 20% to 80% higher than the demagnetized values. The variability of the intensity is undoubtedly related to changes in the nature, composition and quantity of magnetic carriers incorporated in the sediment. In general, the declination and inclination values reflect a normal dipole configuration of the earth's magnetic field. A notable exception is found in the top 100 cm of undisturbed sediments. Samples 2, 3 and 5 near the top of the Teleki Valley profile show persistent intermediate polarity with progressive stepwise demagnetization. Another six samples below sample 5 show a reversed

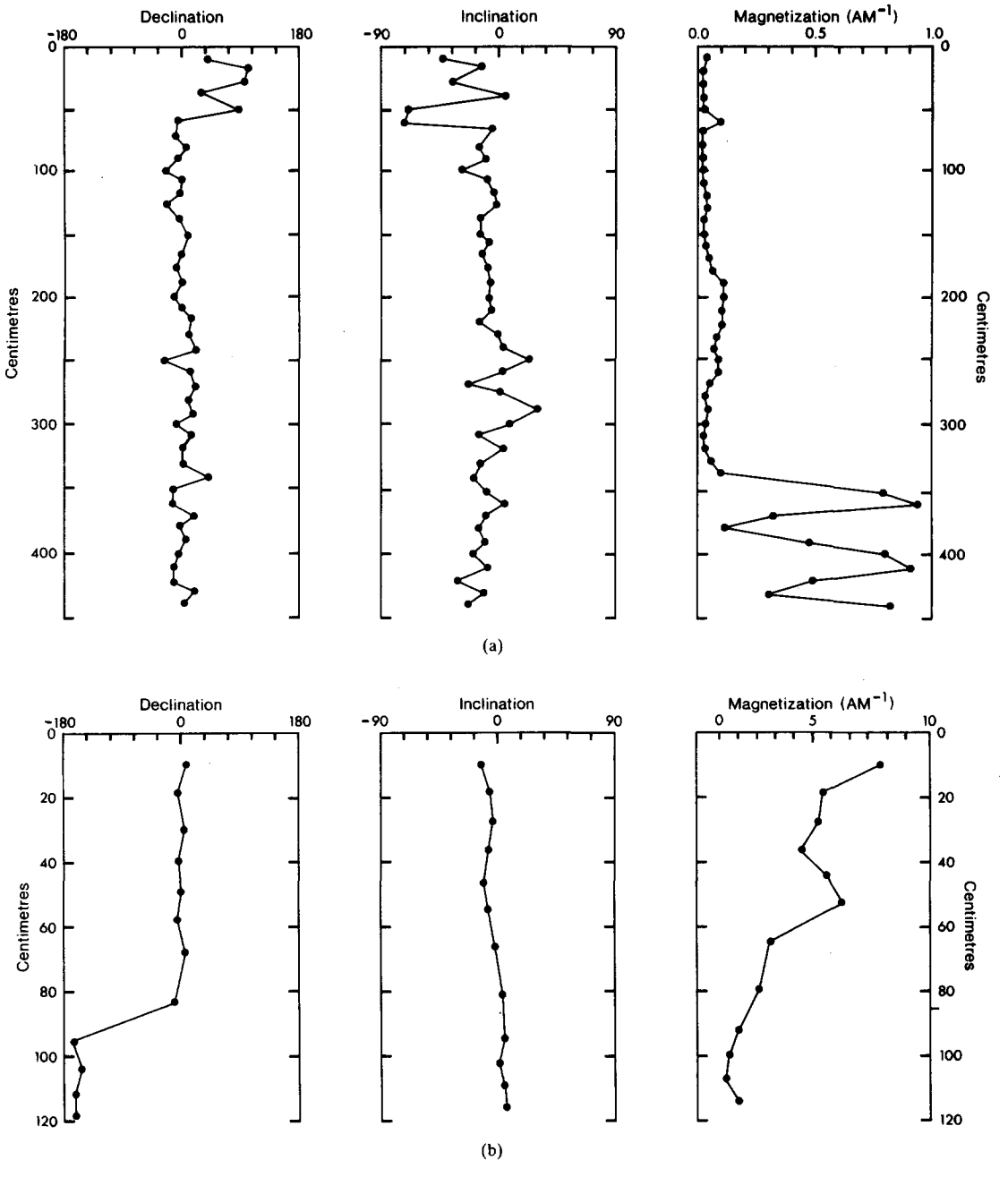


Fig. 5a. Magnetization profiles for site TV61, showing declination, inclination and intensity values for all samples at 15 mT demagnetization.

Fig. 5b. Magnetization profiles for site HOB19, showing declination, inclination and intensity values for all samples at 15 mT demagnetization.

Table 1. Summary of remanent magnetization directions and mean "A" pole positions (for definition of "A" pole, see Irving, 1964)

Symbol	N	M	D	I	K	$\infty 95$	"A" Pole Lat.	"A" Pole Long.
TV61 (Normal)	35	0.16	1	−5	23	9	88°N	199°
TV61 (Excursion)	9	0.15	158	−1	3	35	68°S	130°
HOB19 (Normal)	8	5.11	358	−23	132	9	78°N	225°
HOB19 (Reversed)	4	0.85	188	−7	221	6	81°S	284°
HOB19 (Normal Overprint)	4	2.02	309	4	111	9	39°N	310°

N = number of samples per site.
M = average intensity of magnetization (AM^{-1}).
D = average declination (degree).
I = average inclination (degrees).
K = precision parameter.
$\infty 95$ = radius of the 95% confidence circle about the mean direction (in degrees).

polarity before demagnetization. It is thought that samples 2, 3 and 5 record a short-lived excursion that would have affected sediments laid down during this time. Samples 6 to 11 may represent sediments deposited previously but which nevertheless received a weak reversed overprint that is completely removed at 15 to 25 mT demagnetization (see sample 6, Fig. 3). Perhaps this overprint occurred as a result of active pedogenic or hydrologic conditions which were present at the time and were conducive to remineralization of the remanence carriers. Radiocarbon dates obtained from whole-soil organic materials contained in a zone (IIA11b and IIA12b horizons) just below the area from which paleomagnetic samples 1–11 were collected gives dates of 21,510 ± 430 (Beta–11371) and 24,920 ± 450 (Beta–11372) yr B.P. We suggest that the younger date may be from sediments which were contaminated by younger ^{14}C moving downward from the surface soil profile.

Short-lived excursions between 20,000 and 25,000 yr B.P. have been proposed for deposits in France, Denmark, Australia, Iceland, Canada and U.S.A. (Tarling, 1983; Verosub, 1982 and 1986). Whether or not this excursion correlates with any of those found in other countries is difficult to ascertain at present. Further suitable sediments of this age range will have to be found to determine its areal extent in East Africa.

Hobley Valley site

The declination, inclination, and intensity profiles for all samples are shown in Fig. 5b and the summary of remanent magnetization directions is shown in Table 1. The intensities of the demagnetized samples range from 0.6 to 9.0 AM^{-1}. Undemagnetized intensities are from 70% to 90% higher than the demagnetized values. As the deposits sampled are largely till and loess, some variability in nature, composition, and quantity of magnetic carriers is to be expected. The inferred polarity of the Hobley Valley sediments shows a normally magnetized upper zone (loess) and a reversely magnetized lower zone (till). The change from normal to reversely magnetized sediments is associated with an abrupt shift in declination of nearly 180°, no major change in inclination, and a significant weakening of magnetic moment (Fig. 5b). These changes are typical of magnetic field behaviour at equatorial latitudes, during a reversal. Transition polarities are not seen, possibly indicating a hiatus between the deposition of the reversely magnetized till and the normally magnetized loess.

To estimate the timing and age of these two polarity events, it is necessary to take into consideration the Quaternary stratigraphic and morphologic evidence of multiple glacial events on Mount Kenya, and the sense of the polarity reversal (with decreasing age, the sequence changes from reversed to normal polarity). The till found at Hobley Valley site HOB19 is at an elevation of approximately 3150 m above M.S.L. and marks the lowermost advance of ice down the southeast flank of Mount Kenya. These oldest glacial deposits have been labelled pre-Teleki (Mahaney, 1979, 1983). It is assumed on Mount Kenya, as elsewhere, that the strongest glaciations were of early Pleistocene age, and that these glaciers would have deposited the moraines and associated tills found at lower elevations. The earliest glacial deposits which are recognizable on Mount Kenya have been described by Nilsson (1931), Baker (1967) and Mahaney (1979). At one location, sediments thought to be glacial in origin, are overlain by basaltic tuff (Baker, 1967). Correlation with adjacent volcanic deposits, imply an early Pleistocene age for this tuff. On the basis of this information, and the reversely magnetized till found in Hobley Valley an age of approximately 730,000 yr B.P. or greater can be assigned to the oldest tills found on Mount Kenya. The magnetic record for HOB19 may mark the Brunhes/Matuyama border (0.73 m.y.) or one of the magnetic events within the Matuyama Chron. Ages of 0.9 and 0.97 m.y. for the Jaramillo, 1.87 m.y. for the Olduvai and 2.04 and 2.12 m.y. for the Réunion are also likely candidates in that they mark polarity transitions from a reversed to a normal field with decreasing time during the Pleistocene. There can be no doubt that the Hobley Valley record marks one of these early Pleistocene or late Pliocene magnetic boundaries. With additional work at these and other sites, the glacial sediments on Mount Kenya offer considerable potential for paleomagnetic correlation and dating.

CONCLUSIONS

Profile TV61 consists of a complex Paleudalf formed in till and outwash that predates the last glaciation. While the lower half of the profile gives a normal polarity, it may well fall within the subchrons of the Matuyama (*e.g.* Jaramillo, Olduvai or Réunion) that give normal signatures. A high ratio of D to L aspartic acid, very fine soil textures, and higher degree of weathering indicate that the lower part of the profile is considerably older than the radiocarbon ages suggested for the upper half of the profile. Sediments in the uppermost part of the profile record a possible magnetic excursion and have yielded radiocarbon dates of around 25,000 yr B.P. Further work will be needed before the certainty of this excursion can be established.

Profile HOB19, a Paleudalf, formed in pre-Pleniglacial till is magnetically reversed suggesting that some of these older tills are older than the Brunhes/Mutuyama boundary (~0.7×10^6 yr B.P.). Just where this particular till lies relative to the Jaramillo subchron (0.90–0.97×10^6 yr B.P.) is the subject of future research.

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